

PROJECT PROPOSAL

ICT 212-2

Location-Based Mobile Marketplace Connecting Skilled Workers and Customers

Group Number 12

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1.0 Introduction

1.1 Project Title

Location-Based Mobile Marketplace Connecting Skilled Workers and Customers

1.2 Project Description

Many customers need quick help from nearby skilled workers such as plumbers, carpenters, electricians, painters, masons, mobile repair technicians, and other small-scale service providers. However, many individual workers do not have websites, social media pages, Google Business profiles, or a proper digital presence. Therefore, customers often depend on personal contacts, roadside signboards, or word-of-mouth recommendations when they need urgent or reliable service support.

The proposed project is a mobile-based marketplace that connects customers with nearby skilled service providers using location-based search, service categories, verified provider profiles, request management, ratings, reviews, and notifications. Customers can search for service providers, view profile details, submit a service request with a problem description and images, track the request status, and rate the completed service. Service providers can register, create their profiles, choose service categories and working areas, receive customer requests, accept or reject jobs, update job status, and view job history.

The project focuses mainly on individual and small-scale skilled workers rather than established shops, because shops are usually easier to find through map applications while independent workers are often less visible online. The proposed system also includes an administrator web panel to verify providers, manage users, maintain service categories and locations, monitor service requests, handle complaints, and view basic platform reports.

The project includes two practical AI-assisted features. First, a service category suggestion feature allows a customer to describe a problem in natural language and receive a suitable service category suggestion. Second, a request description generator helps customers convert short or unclear problem statements into clear service request descriptions before submission. These AI features are designed to support the marketplace workflow without making AI the main system.

2.0 Background and Motivation

2.1 Background of the Problem

In many local communities, finding skilled workers for small jobs remains informal and inefficient. Customers may know a few workers personally, but they may not know whether those workers are currently available, nearby, trustworthy, or suitable for the required service. This creates delays and uncertainty, especially when the service requirement is urgent or when the customer is new to the area.

At the same time, small-scale skilled workers often face a lack of digital visibility. A Sri Lankan MSME digital tool usage study found that digital tools became important for business continuity, but many MSMEs still faced difficulties such as poor internet connectivity and lack of knowledge when using digital tools [12]. Sri Lanka's National Digital Strategy also highlights digital transformation as a way to support economic growth, employment creation, sustainable development, and service delivery [13]. Therefore, a location-based digital marketplace can reduce the gap between customers and skilled workers by making workers easier to discover and by giving customers a structured way to request and track services. In many local communities, finding skilled workers for small jobs remains informal and inefficient. Customers may know a few workers personally, but they may not know whether those workers are currently available, nearby,

trustworthy, or suitable for the required service. This creates delays and uncertainty, especially when the service requirement is urgent or when the customer is new to the area.

Many skilled people depend only on personal recommendations and local contacts to receive work. This can reduce their job opportunities even when there are customers nearby who need their skills. A location-based digital marketplace can reduce this gap by making skilled workers easier to discover and by giving customers a structured way to request and track services.

2.2 Previous Research and Existing Approaches

Several existing methods help customers find services, but they do not fully solve the problem of discovering individual small-scale skilled workers through a controlled request workflow. Table 1 summarizes the main existing methods and their limitations.

Table 1: Existing systems and Limitations

Existing Method/System	Features	Limitations
Google Maps / map search	Find registered shops and businesses by location.	Google Business Profile helps storefronts or service-area businesses appear on Google Search and Maps, but providers must add, claim, and verify business information. Individual informal workers may not always maintain such verified searchable profiles [14], [15].
Facebook groups / WhatsApp contacts	Customers can ask community members for recommendations.	Information is scattered, difficult to verify, and not organized by category, distance, or availability.
Classified advertisement websites	Service providers can post service advertisements.	Usually work as listing platforms and may not support verified profiles, request tracking, or ratings.
General home-service apps	Support selected services with booking features.	May focus on larger cities, formal service businesses, or selected categories rather than informal local workers.
Direct phone contact	Fast when the customer already knows a provider.	No discovery support, no request history, no rating mechanism, and no platform-level trust features.

Trust is a major issue in online service marketplaces because customers often make decisions without personally knowing the service provider. Previous studies show that reputation systems, ratings, and reviews help users decide whether to transact with a provider and encourage better service behavior [16], [17].

2.3 State of the Art

Modern service marketplaces commonly use secure authentication, role-based access, digital provider profiles, category-based browsing, location-aware search, ratings, reviews, notifications, and administrator dashboards. Location-based services are useful in mobile commerce because they allow systems to discover relevant services based on the user's current or selected location [18]. In online marketplaces, trust and reputation mechanisms such as ratings, reviews, and profile information help reduce uncertainty between buyers and service providers [16], [17]. Local and regional digital platforms are also important because they can respond to local market needs, support local workers, and reduce entry barriers compared with large global platforms [19].

The proposed system adapts these modern marketplace concepts to a local small-scale service environment. Instead of functioning only as a directory, the system uses a controlled request workflow where the customer sends a request through the application and the provider responds through the platform. This helps the system maintain request history, status tracking, ratings, and future monetization opportunities.

2.4 Motivation

The motivation for this project comes from the practical difficulty of finding reliable skilled workers quickly and the lack of online visibility for small-scale workers. The system can support local communities by improving access to trusted services and can support skilled workers by creating a digital channel for receiving job requests. For the development team, the project provides a strong opportunity to apply mobile application development, cloud databases, geolocation, AI integration, authentication, software design, and system testing to a real-world problem.

3.0 Problem

3.1 Problem Statement

Customers face difficulty when searching for nearby skilled workers because information about individual service providers is often unavailable, scattered, outdated, or not verified. Existing tools such as Google Maps are useful for finding listed businesses, but searchable visibility usually depends on businesses adding, claiming, and verifying their profiles [14], [15]. Skilled workers also face a visibility problem because many of them do not have a proper online presence, and MSME studies show that digital tool adoption can still be limited by connectivity and knowledge barriers [12].

3.2 Importance of the Problem

Solving this problem is important because it benefits both customers and skilled workers. Customers can save time, reduce uncertainty, and find appropriate help more easily. Service providers can increase their visibility, improve trust through profile verification and reviews, and receive more job opportunities. The system also supports digital inclusion by helping small-scale workers participate in the digital service economy.

3.3 Proposed Solution

The proposed solution is a location-based mobile marketplace connecting skilled workers and customers. For the MVP, the system will be implemented as one Android mobile application with role-based interfaces for customers and service providers. This avoids the complexity of developing two completely separate mobile apps while still providing different features for each user role. A separate web-based administrator panel will be developed for platform management.

The system will allow customers to register, log in, search nearby providers by category and location, and view provider profiles. An AI-based feature will assist users in selecting the most suitable category and improving the clarity of request descriptions. Separately, the application will allow users to submit requests with images and location details, track request status, and rate completed jobs. Providers will be able to create profiles, select categories and service areas, upload verification documents, manage availability, handle incoming requests, update job status, and view job history. Administrators will verify providers, manage users and categories, monitor requests, manage complaints, and view basic reports.

The platform will not mainly expose provider contact information before a request is created. Instead, customers submit a request through the app and the provider accepts or rejects it. After acceptance, limited contact or communication details may be enabled if required. This approach helps maintain platform value, protects privacy, and allows the system to keep request history, status tracking, and review records.

3.4 Use Case Diagram

The main use case diagram in Figure 1 uses four actors: Customer, Service Provider, Administrator, and AI Assistant Service. The AI Assistant Service is shown as an external supporting actor only for service category suggestion and request description generation.

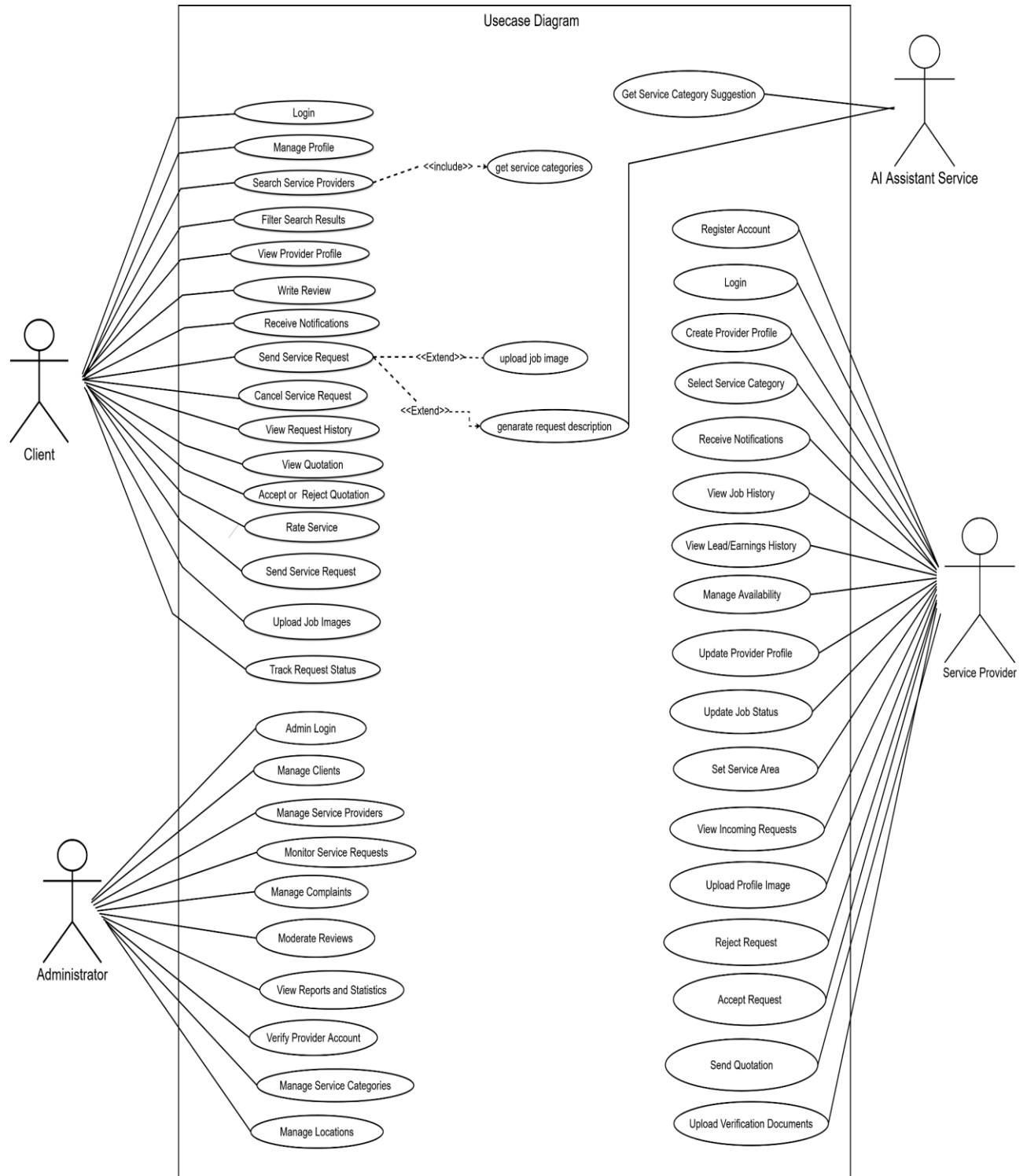


Figure 1: Use Case Diagram

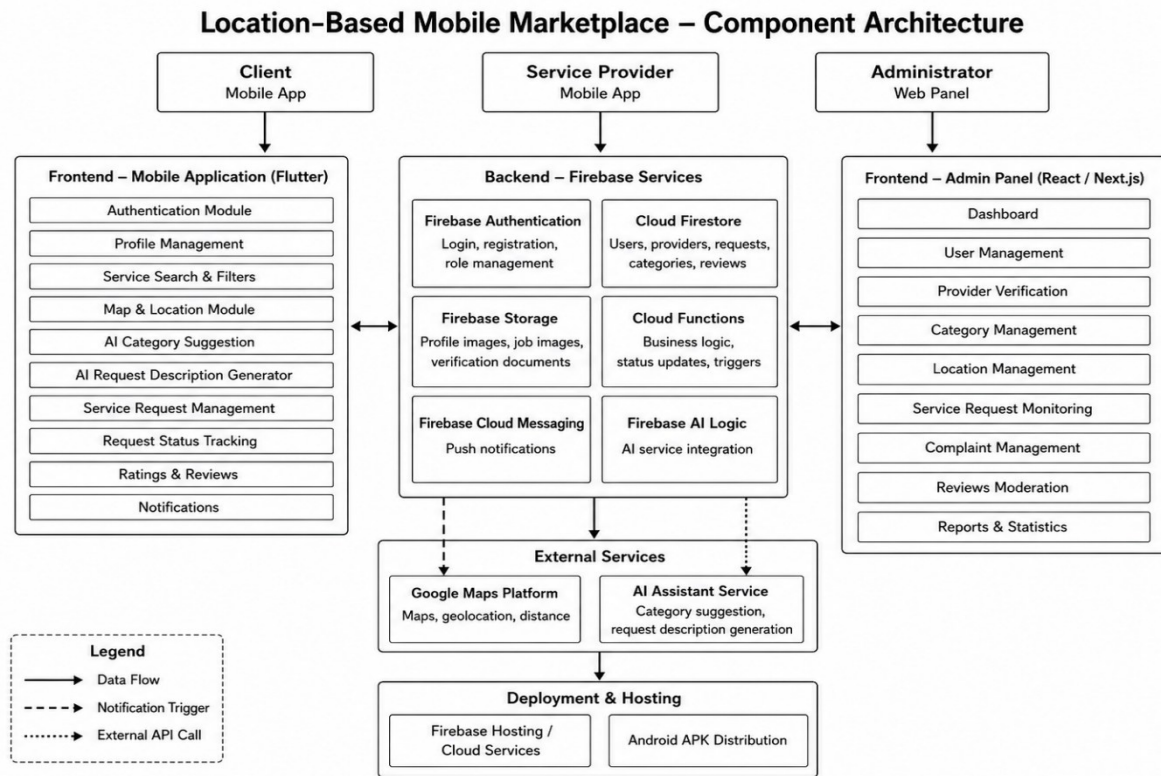


Figure 2: Component Architecture

4.0 Aim and Objectives

4.1 Aim

The aim of this project is to develop a location-based mobile marketplace system for connecting customers with nearby skilled workers using mobile application development, cloud database services, geolocation, role-based access control, provider verification, and selected AI-assisted features.

4.2 Objectives

- To develop an Android mobile application with role-based interfaces for customers and service providers.
- To implement secure registration, login, and profile management for customers and service providers.
- To implement location-based service provider search using category, location, distance, availability, and rating information.
- To allow customers to submit service requests with descriptions, location information, preferred time, and optional images.
- To allow service providers to receive, accept, reject, and update service request statuses.
- To implement provider verification, ratings, reviews, and complaint handling to improve platform trust.
- To implement two AI-assisted features: service category suggestion and request description generation.
- To develop an administrator web panel for managing users, providers, categories, locations, complaints, and reports.
- To design the system so that future monetization features such as platform fees, subscriptions, or promoted listings can be added.

5.0 Methodology

5.1 Software Development Methodology

This project will follow an iterative software development methodology. The system will be developed in multiple iterations, where each iteration focuses on planning, design, implementation, testing, and refinement of selected features. This approach is suitable because it allows the team to first develop the core system functionalities and then progressively improve the system based on feedback from supervisors, users, and testing activities.

The main iterations will include requirement analysis and research, system design and UI planning, authentication and profile development, provider search and map integration, AI feature implementation, service request workflow, notification support, administrator panel development, integration testing, final documentation, and presentation preparation.

5.2 Technology Stack

The proposed technology stack is selected to support fast development, cloud scalability, mobile-first implementation, and practical AI integration. Flutter is selected for mobile application development because it supports building applications from a single codebase for multiple platforms [1]. Firebase Authentication is used to provide secure registration and login for customers, service providers, and administrators [2]. Cloud Firestore is selected as the main cloud database for storing user profiles, provider profiles, service categories, service requests, quotations, ratings, reviews, and status updates [3]. Firebase Storage is used to store profile images, service request images, and provider verification documents [4]. Firebase Cloud Messaging is used to send notifications for request updates, quotation activities, job status updates, and provider verification results [5]. Firebase AI Logic / Gemini API is used to support the AI-assisted service category suggestion and request description generation features [6]. Google Maps Platform and device geolocation APIs are selected to support map display, location selection, nearby provider search, and approximate distance calculation [7]. React is selected for the administrator web panel because it supports component-based user interface development [8]. GitHub will be used for source-code management, version control, and team collaboration [9].

Table 2: Proposed Technology Stack

Layer/Module	Technology	Purpose
Mobile application	Flutter and Dart	Develop one Android application with role-based customer and service provider screens.
Admin web panel	React	Build a dashboard for provider verification, category management, complaints, and reports.
Authentication	Firebase Authentication	Provide secure registration and login for customers, service providers, and administrators.
Database	Cloud Firestore	Store user profiles, provider profiles, categories, service requests, quotations, ratings, reviews, and status updates.
File storage	Firebase Storage	Store profile images, job images, and provider verification documents.
Notifications	Firebase Cloud Messaging	Send notifications for request updates, provider responses, and verification results.
AI features	Firebase AI Logic / Gemini API	Support service category suggestion and request description generation.
Maps and location	Google Maps Platform and device geolocation APIs	Support current location capture, map location selection, nearby search, and approximate distance.
Version control	Git and GitHub	Support source-code management and team collaboration.
Testing	Manual testing, Flutter widget tests, and selected unit tests	Validate role-based flows, search, AI assistance, request status updates, and admin functions.

5.3 Functional Requirements

5.3.1 Customer Functional Requirements

CFR-01 Register Account

The system shall allow users to create an account using required personal details.

CFR-02 Login

The system shall allow registered users to securely log in and access their dashboard.

CFR-03 Manage Profile

The system shall allow users to view and update their profile information.

CFR-04 Search Service Providers

The system shall allow users to search for service providers based on category and location.

CFR-05 Get Service Category Suggestion

The system shall provide AI-based suggestions to help users identify the appropriate service category.

CFR-06 View Provider Profile

The system shall allow users to view detailed information about service providers.

CFR-07 Send Service Request

The system shall allow users to submit service requests to selected providers.

CFR-08 Generate Request Description

The system shall allow users to generate or improve service request descriptions using AI assistance.

CFR-09 View Quotations

The system shall allow customers to view quotations sent by service providers, including estimated service cost, inspection fee (if applicable), material cost notes, available time, and provider message.

CFR-10 Accept or Reject Quotation

The system shall allow customers to accept or reject a quotation before the service request is confirmed.

CFR-11 View Quotation Status

The system shall allow customers to view quotation status such as pending, received, accepted, rejected, or expired.

CFR-12 Track and Cancel Request

The system shall allow users to track the status of service requests and cancel requests when allowed.

CFR-13 Rate and Review Service

The system shall allow users to rate and review completed services.

CFR-14 View Request History

The system shall allow users to view their past and current service requests.

5.3.2 Service Provider Functional Requirements

SPFR-01 Register Account

The system shall allow service providers to create an account by providing required professional and contact information.

SPFR-02 Login

The system shall allow registered service providers to securely log in and access their dashboard.

SPFR-03 Manage Profile

The system shall allow service providers to create and update their profile information.

SPFR-04 Upload Verification Documents

The system shall allow service providers to upload verification documents for administrative review.

SPFR-05 Manage Availability

The system shall allow service providers to set and update their availability status.

SPFR-06 Receive and Manage Service Requests

The system shall allow providers to receive service requests, send quotations, handle accepted quotations, reject unsuitable requests, and update job status.

SPFR-07 Review Customer Requests

The system shall allow service providers to review customer service requests before accepting a job.

SPFR-08 Send Quotation

The system shall allow service providers to send a quotation for a received request, including estimated service charge, inspection fee, material cost note, available time, and additional message.

SPFR-09 Update Job Status

The system shall allow service providers to update the status of assigned jobs.

SPFR-10 View Job History

The system shall allow service providers to view their past service request history.

SPFR-11 View Performance Overview

The system shall allow service providers to view a summary of their service activity and performance metrics.

5.3.3 Administrator Functional Requirements

AFR-01 Admin Login

The system shall allow authorized administrators to securely log in to the admin panel.

AFR-02 Manage Users

The system shall allow administrators to manage customer and service provider accounts.

AFR-03 Verify Providers

The system shall allow administrators to review and verify service provider accounts and submitted documents.

AFR-04 Manage Service Categories and Locations

The system shall allow administrators to manage service categories and supported service locations.

AFR-05 Monitor Service Operations

The system shall allow administrators to monitor service requests, quotations, final job records, and complaints.

AFR-06 Manage Complaints

The system shall allow administrators to manage and respond to user complaints.

AFR-07 Moderate Reviews

The system shall allow administrators to moderate user-generated reviews.

AFR-08 View Reports

The system shall allow administrators to view summary reports of system activity and usage statistics.

5.3.4 System Functional Requirements

SFR-01 Android Support

The system shall support Android devices for the mobile application.

SFR-02 Role-Based Access Control

The system shall provide role-based access to features based on user type.

SFR-03 Data Management

The system shall store and manage user data, provider data, service requests, quotations, ratings, reviews, categories, and location information.

SFR-04 Location Handling

The system shall support capturing and selecting user locations for service requests and enable distance-based service matching.

SFR-05 Image Support

The system shall allow users to upload images for profiles, service requests, and verification purposes.

SFR-06 Notification Service

The system shall notify users about important system events such as request updates, verification results, and quotation activities.

SFR-07 AI Assistance Service

The system shall provide AI-based assistance for service category suggestion and service request description improvement.

SFR-08 Quotation Status Management

The system shall maintain quotation statuses such as pending, quoted, accepted, rejected, confirmed, in progress, completed, and cancelled.

SFR-09 Quotation Data Storage

The system shall store quotation details and final job amount securely in the cloud database.

SFR-10 Request Status Management

The system shall maintain and synchronize service request statuses across all user roles.

5.4 Non-Functional Requirements

5.4.1 Performance Requirements

- The system shall display nearby provider search results within 5 seconds under normal internet conditions.
- The mobile application shall load main screens and role-based dashboards within 5 seconds on Android 8 or newer devices.
- The system shall compress uploaded images before storage to reduce bandwidth usage and improve upload performance.

5.4.2 Security Requirements

- The system shall require authenticated login before allowing users to access role-specific features or personal data.
- The system shall enforce role-based access control for customers, service providers, and administrators.
- The system shall protect user data, request details, images, and verification documents using Firebase Security Rules and authorized access.
- The system shall validate required inputs and uploaded files before processing or storing data.

5.4.3 Usability Requirements

- The system shall provide a simple and consistent interface suitable for customers, service providers, and administrators with basic digital skills.
- The system shall guide service providers through registration, profile completion, category selection, and verification document upload.
- The system shall display clear messages for request, quotation, verification, complaint, and job status changes.
- The system shall reduce unnecessary typing through service categories, dropdown lists, location selection, and AI-assisted description support.

5.4.4 Reliability Requirements

- The system shall store important records such as users, provider profiles, service requests, quotations, reviews, and complaints in the cloud database.
- The system shall notify users when internet connectivity problems prevent data loading, submission, or synchronization.
- The system shall maintain consistent request, quotation, and job status information across customer, provider, and administrator interfaces.

5.4.5 Scalability Requirements

- The system architecture shall support adding more service categories, locations, users, and service requests as platform usage increases.
- The system shall allow future features such as in-app chat, online payments, subscriptions, and promoted listings without major redesign.

5.4.6 Maintainability Requirements

- The system shall be developed using separate modules for authentication, profiles, search, requests, quotations, notifications, AI features, and administration.
- The system shall use reusable UI components and clear coding practices to simplify future updates and bug fixes.
- The development team shall use Git and GitHub for version control, collaboration, and source-code backup.

5.4.7 AI Requirements

- The system shall allow users to review and edit AI-generated category suggestions and request descriptions before submission.
- The AI features shall be limited to service category suggestion and request description improvement within the project scope.
- The AI features shall not make final service decisions, approve providers, assign jobs, or replace administrator review

6.0 Resource Requirements and Budget

6.1 Hardware Requirements

Table 3: Hardware Requirements

Resource	Minimum/Recommended Requirement	Purpose
Developer laptop/desktop	Intel i3/Ryzen 3 or higher, 8 GB RAM recommended, SSD preferred	Development, testing, documentation, and admin panel implementation.
Android smartphone	Android 8 or newer, GPS, camera, stable internet	Testing mobile app, location features, image upload, and notifications.
Admin testing computer	Modern browser, 4 GB RAM minimum	Testing administrator web panel.
Internet connection	Stable broadband or mobile data	Development, cloud services, testing, maps, and AI integration.

6.2 Software Requirements

Operating System:

Windows 10/11 or Linux

Development Tools (IDE):

Visual Studio Code, Android Studio

Mobile Development Framework:

Flutter SDK, Dart

Admin Panel Development:

React

Backend & Cloud Services:

Firebase Authentication, Cloud Firestore, Firebase Storage, Firebase Cloud Messaging, Firebase Security Rules, and optional Firebase Cloud Functions

AI Services:

Firebase AI Logic or Gemini API

Maps & Location Services:

Google Maps Platform, Geolocation packages

Version Control System:

Git, GitHub

Design Tools:

Figma, draw.io (or similar diagram tools)

Testing Tools:

Android Emulator, physical Android devices, browser developer tools, unit/widget testing tools

6.3 Estimated Budget

Most selected development tools and Firebase services can be used within no-cost or free-tier limits during MVP development. However, future costs may increase depending on Firebase usage, Google Maps Platform API usage, storage, notifications, and AI feature requests [10], [11].

Table 4: Estimated budget

Item	Description	Estimated Cost
Flutter, Dart, GitHub, VS Code	Free tools used for mobile app development and version control.	0 LKR
Firebase Authentication	Used for Google login and email/password login for customers and service providers.	0 LKR initially (Cost may increase with higher usage)
Cloud Firestore Database	Stores user details, worker profiles, service requests, ratings, and request status.	0 LKR initially (Cost may increase with higher usage)
Firebase Cloud Storage	Stores profile images and service-related images. Cost will remain low if images are compressed.	0 LKR initially (Cost may increase with higher usage)
Firebase Cloud Messaging	Used to send notifications for service requests and status updates.	0 LKR
Cloud Functions / Backend Logic	Used for secure backend operations such as notifications and AI API calls.	0 LKR initially (Cost may increase with higher usage)
Firebase AI Logic / Gemini API	Used for AI-based category suggestions and service request description support.	0 LKR initially (Cost may increase with higher usage)
Google Maps Platform	Used for map display, location selection, and nearby worker search. Free credits/limits can be used during development.	0 LKR initially (Cost may increase with higher usage)
Domain Name	Optional, only needed for a future website or admin panel.	3,000–7,000 LKR/year

7.0 Project Time Plan

The proposed time plan is designed for a 15-week university module. The plan begins with requirement analysis and design, then moves through iterative development, integration, testing, documentation, and final presentation.

Table 5: Iteration Plan and Deliverables

Iteration	Weeks	Main Focus	Output
Iteration 1	Week 1 – Week 3	Requirement analysis, research, system design, UI planning	Finalized requirements, diagrams, UI wireframes
Iteration 2	Week 3 – Week 5	Authentication, role-based access, customer/provider profiles	Login, registration, dashboards, profile modules
Iteration 3	Week 6 – Week 8	Location-based search, maps, AI features	Nearby provider search, category suggestion, request description generator
Iteration 4	Week 9 – Week 11	Service request, quotation, job status, notifications, ratings	Request → quotation → approval → completion workflow
Iteration 5	Week 11 – Week 15	Admin panel, integration testing, bug fixing, documentation, presentation	Complete MVP, final report, final demo

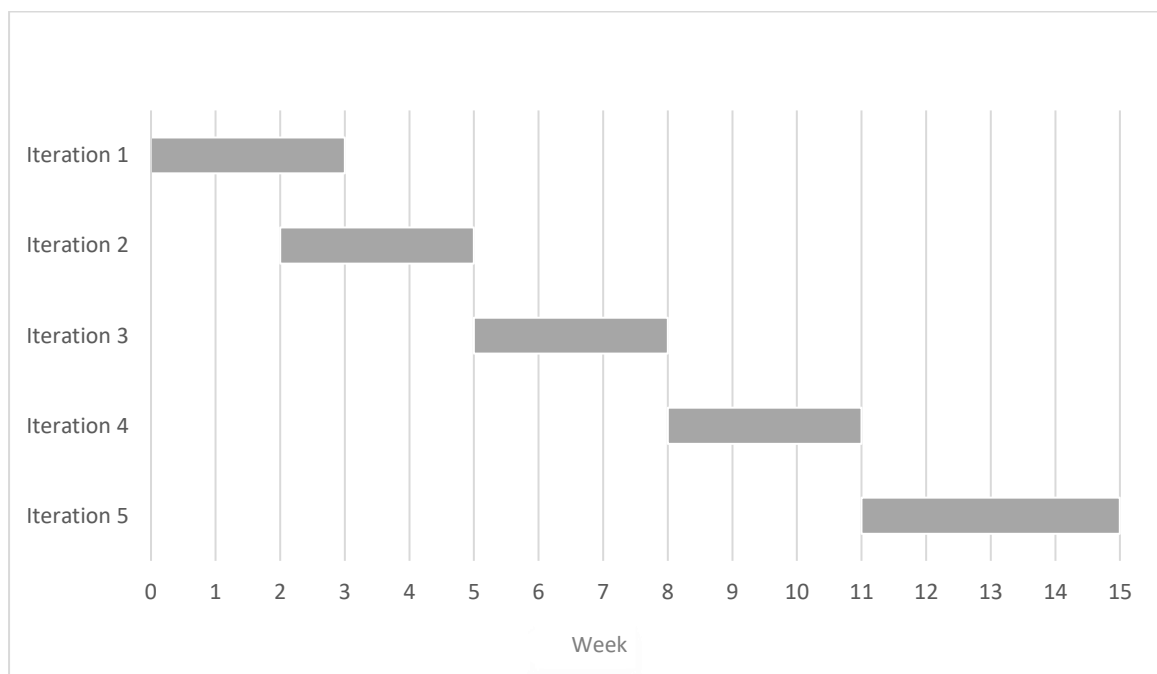


Figure 3: Gantt Chart

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9.0 Appendices

Appendix A: System Scope

The proposed system includes customer and service provider registration and login with role-based access control. Customers can search for nearby service providers based on category and location, view provider profiles, submit service requests, receive quotations, track request progress, and provide ratings and reviews after service completion.

Service providers can create and manage professional profiles, upload verification documents, review customer requests, send quotations, manage job statuses, and maintain service availability.

The system also includes AI-assisted features such as service category suggestions and request description generation to improve usability and request accuracy.

Administrative functions include provider verification, user and category management, complaint handling, review moderation, quotation and request monitoring, and viewing system activity reports.

Additional system support features include notifications, cloud-based data storage, image uploads, location handling, and status synchronization across all user roles.

Appendix B: Future Enhancements

- In-app chat between customers and service providers.
- Online payment integration and commission handling.
- Advanced provider recommendation based on distance, rating, availability, and completed jobs.
- Multilingual support for Sinhala, Tamil, and English.
- Mobile app release for iOS after Android MVP validation.

10.0 Team Member Signatures

Registration Number	Name	Contact Number	Signature
UWU/ICT/23/006	D.S.R. Dissanayake	076-7092322	
UWU/ICT/23/002	M.T. Kaushalya	070-4592008	
UWU/ICT/23/064	K.D.P. Kandegama	076-9727678	
UWU/ICT/23/073	L.R.N.P. Rajakaruna	076-1980159	
UWU/ICT/23/021	W.M.D.H. Chathumina	077-0781634	

Date: 16/05/2026

Recommendation of the supervisor(s)

Name: _____

Department/Organization: _____

Signature: _____